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End Semester Examination June 2024-25

Name of the Course: B.Tech
(BioTech)

Semester: **VI**

Name of the Paper: Fundamental of
Statistics and AI

Paper Code: TCS 421

Time: 3 Hours

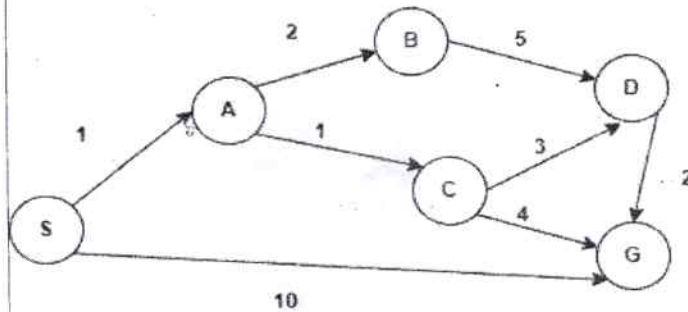
Maximum Marks: 100

Note: -

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question
- Total marks in each main question are twenty.
- Each question carries 10 marks

Q1	(20 marks)	CO1,
(a)	Explain different types of AI with examples.	CO2
(b)	Illustrate Intelligent Agents. Explain the different types of Agents.	
(c)	Compare Depth Limited and Depth First Search Method. Find the path to reach the goal node (4) from the star node (0) by applying DFS.	
Q2	(20 marks)	CO2
(a)	What are the five key elements involved in formulating a problem for a problem-solving agent? In what ways can such agents approach and solve problems?	
(b)	Describe the foundation and current trends of AI in detail.	
(c)	Explain the advantages and limitations of the Breadth-First Search (BFS) algorithm, and use it to traverse the given graph starting from node A and goal node C.	
Q3	(20 marks)	CO2, CO3
(a)	Describe the importance of Exploratory Data Analysis and outline the key steps involved in performing EDA.	
(b)	Discuss A* search algorithm and apply the same for the given graph to search the goal node. Consider Start Node=S and Goal Node=G.	

S	A	B	C	D	G
5	3	4	2	6	0



(c)	Compare supervised and unsupervised learning and illustrate the differences using examples.	
Q4		(20 marks)
(a)	How does data pre-processing differ from data cleaning? What are the main steps involved in data pre-processing, and why is it a crucial part of data analysis?	CO3, CO4
(b)	Compare data science and business intelligence in terms of goals and methods. In what ways can business intelligence support data-driven decision-making?	
(c)	Explain barplot, linechart using R programming with the help of an example.	
Q5		(20 marks)
(a)	What is the Pearson correlation coefficient? Calculate the mean, covariance, and Pearson correlation coefficient between the following variables: • $X = [1, 2, 3, 4, 5]$ • $Y = [2, 4, 6, 8, 10]$ Make a scatter plot using python or R programming.	CO5
(b)	What is linear regression, and how is it different from logistic regression?	
(c)	Write short notes on (i) Populations and Samples (ii) Functions in R Programming (iii) Regression	